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years. Second, he characterizes these theories in terms of the relative weights each assigns to rules and to representations. His aim is to demonstrate that adequate theories cannot rely exclusively on one or the other; some sort of balance is essential. Third, Anderson examines the influences of theories and theoreticians on each other, and compares theories, especially with regard to their reliance on representations and rules. Finally, and by no means of least importance, the author provides biographical data concerning the major theoreticians, and places them (and their theories) in relief against the background of linguistic and general intellectual thought of the times in which they worked. The evenhandedness of Anderson's analyses and evaluations of widely divergent theories, combined with his satisfyingly comprehensive treatment, make this book essential reading for linguists.—*Lawrence J. Raphael, Speech, Herbert H. Lehman College*

Animal Bone Archeology: From Objectives to Analysis. Brian Hesse and Paula Wapnish. Manuals on Archeology, 5. 132 pp. Taraxacum, 1985. $18.

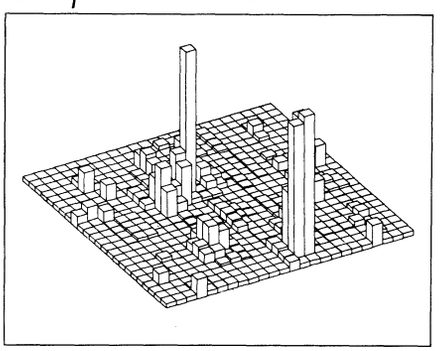
This is the first zooarchaeology manual to place primary emphasis on the importance of having cultural and historical questions guide faunal analysis. As such, it is a welcome addition to the growing body of literature on zooarchaeological theory and method. The first two chapters make clear the potential of faunal data for addressing questions about economy, social organization, and ideological systems at all levels of human sociocultural development. Chapters 3 through 8 focus on the tools which allow faunal data to be used in this way. Issues include processes responsible for meaningful patterning in the data and those which distort these patterns. Formulation of sampling, collection, recording, and storage protocol are discussed. Analysis techniques are presented as they apply to the study of individual bones, specific taxon or groups of bones, and examination of the entire assemblage. Most discussions are comprehensive and clear, with ample bibliographic references, photographs, and drawings.

Discussions of analytical methods focus primarily on distinguishing between noncultural and cultural patterning in faunal data. Although recognition of the impact of noncultural bias is essential, the authors place too much emphasis on identification of bias and too little on identification of meaningful patterning. Also, it is sometimes unclear whether certain techniques are best used to examine sample taphonomy or to monitor human utilization of animals. Moreover, the sequence of discussions from bone to assemblage suggests that research should proceed from data to interpretation, rather than the more problem-oriented approach the authors advocate.

This impression is partially dispelled in the final chapter, where the original emphasis on research questions directing methodological procedures is reiterated. Yet, this cursory summary of preceding chapters fails to articulate the intermediary steps which bridge research problems and specific analytical methods. Moreover, nowhere do the authors address the next step of the research process, in which empirical analysis is related back to research questions, leading to problem clarification, reformulation, and subsequent examination with another set of faunal data.

Nevertheless, this is a fine volume which conveys the great potential of faunal data for the study of cultural evolution, and it provides an impressive quantity of information on how to realize this potential. In fact, even specialists in zooarchaeology would do well to take the theme of this manual to heart and to use it as a guide for their own research.—*Melinda A. Zeder, Anthropology, National Museum of American History, Smithsonian Institution*

Mathematics and Computer Science



The Riemann Zeta-Function: The Theory of the Riemann Zeta-Function with Applications. Aleksandar Ivić. 515 pp. Wiley-Interscience, 1985. $49.95.

The Riemann zeta-function is of central importance in number theory, both as a prototype for all zeta-functions and as a means for obtaining information about particular arithmetic functions. In this book, the author has the latter purpose in mind. After an introductory chapter on classical theorems, he develops the theory of exponent pairs and proves various mean-value theorems and zero density estimates, from which he obtains, as his main applications, refined lower bounds for the number of primes in a short interval and upper bounds for the error terms in some divisor problems. Many of the methods presented here have been developed since Titchmarsh's 1951 treatise; however, this book is far from comprehensive. Some principal omissions are methods which involve Kloosterman sums, methods which assume the Riemann-hypothesis, and theorems on S(T). Overall, the book has a pleasant style, and the lengthy endnotes to the chapters are interesting.—*Brian Conrey, Mathematics, Oklahoma State University, Stillwater*

Ultrametric Calculus: An Introduction to p-adic Analysis. W. H. Schikhof. Studies in Advanced Mathematics, 4. 306 pp. Cambridge University Press, 1984. $39.50.

When a certain natural but unconventional notion of "distance" is introduced on the rational numbers, one obtains a bizarre geometrical object: a totally disconnected topological space in which every triangle is isosceles and any two circles are either disjoint or concentric. This is the realm of p-adic ("ultrametric") analysis. For several decades, number theorists have regarded p-adic analysis as playing a crucial role, often as important as classical real analysis, in many fields. As Weil wrote in the preface to his *Basic Number Theory* (1967): "we see now that the real numbers appear . . . in arithmetical questions . . . as one of the infinitely many completions of the prime field, one which is neither more nor less interesting to the arithmetician than its p-adic companions."

Schikhof's "ultrametric calculus" book is an ambitious and largely successful attempt to provide a thorough foundation of p-adic analysis in a textbook form usable by beginning graduate students and precocious undergraduates. A long list of standard concepts from courses in elementary and advanced calculus, real and complex analysis, are treated systematically in the often dramatically altered form they take in the p-adic world. The explanations are clear and readable, and the numerous well-chosen exercises enhance the book's appeal.

However, one of the book's major strengths—its painstakingly thorough, almost encyclopedic approach—is also a potential weakness from a pedagogical viewpoint. The ratio of general theorems to number-theoretic examples and applications is quite high. Thus, for instance, the reader must wait until page 81 to learn of the Teichmüller representatives—the roots of unity in Z_p. The uninitiated may easily lose sight of the subject's historical origins in concrete questions of interpolation, cyclotomy, representation by quadratic forms. Indeed, interpolation is introduced only on page 98. This should not bother well-motivated students with a high degree of self-discipline, of whom there are many, especially in Europe. But, in my experi-

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